

Assignment for Week 4

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Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(haven)
```

Assignment

1. 14.1.2 (d) (p. 274)
2. 14.2.2 (p. 279)
3. 16.1.5 (a) (p. 323)
4. 16.2.6 (p. 331)
5. 17.2.1 (p. 349)

7) Which of the following datasets are tidy?

```
dta1<- data.frame(year = c(2000:2003,2000:2003),
                  CountryA = c(rep("USA",4),rep("CAN",4) ),
                  CountryB=c(rep("Russia",4),rep("Russia",4) ),
                  Agreement=c(0,1,1,0,0,1,0,0))

dta2<- data.frame(year = c(2000:2003,2000:2003),
                  Dyad = c(rep("USA-RUS",4),rep("USA-RUS",4) ),
                  ConflictType=c(rep("War",4),rep("Minor MID",4) ),
                  ConflictLevel=c(0,3,1,0,0,1,0,0))

dta3<- data.frame(Dyad = c("USA-RUS", "CAN-RUS"),
                  War2002=c(0,2),
                  War2003=c(0,2),
                  War2004=c(0,2),
                  MinorMID2002=c(0,2),
                  MinorMID2003=c(0,2),
                  MinorMID2004=c(0,2))
```

8) Use spread, gather and separate from the tidy package to fix the untidy data.frames from question 7.

9) Correct the common mistakes in the following code snippets

a) Make the following TRUE

```
mean(c("3,000", 2, 1, 0, 4)) == 601.4
```

b) Make the following into a vector TRUE TRUE TRUE

```
EuclidDistancetoKabul <- function(x) {
  sqrt((-64 - x[1]) ^ 2 + (16 - x[2]) ^ 2)
}
x <- EuclidDistancetoKabul(rbind(c(48, 4), c(12, 3), c(80, 13)))
x > 20
```

c) Make the following code install the 'car' package.

```
install.packages(car)
```

d) Correct the following plot command:

```
x <- c(3, 10, 4)
y <- c(2, 3, 4)
df <- data.frame(x, y)
ggplot(df) + geom_bar(aes(x, y))
```

e) Make the following code report TRUE

```
allthetypes <- list(df1 = data.frame(x = c("Iowa"), stringsAsFactors = F))
is.character(allthetypes[1][[1]])
```

- 10) How many shootings occur in the state with the most active VA hospitals? The least active? For this question you will be using the Washington Post shootings data and the VA data described in the R lab for Week 4.
- 11) Plot the following function for $x \in [-4, 4]$ by writing a custom function(). Use `integrate()` on that function to plot the area under the curve of that function from -Inf to x, such that $x \in [-4, 4]$.

$$\frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$